11.4 The Tautochrone

WORKSHEET

The question

A smooth, axially symmetric bowl has its vertical cross-sections determined by $s = 2\sqrt{ky}$, where s is the arc-length measured from its lowest point V, and y is the height above V. A particle is released from rest at a point on the surface at a height h above V. Explain why

$$\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 + 2gy$$

is constant.

Show that the time for the particle to reach V is

$$\pi\sqrt{\frac{k}{2g}}.$$

Two elastic particles of mass m and αm, where $\alpha < 1$, are released simultaneously from opposite sides of the bowl at heights $\alpha^2 h$ and h respectively. If the coefficient of restitution between the particles is α, describe the subsequent motion.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The bowl is smooth. Which forces act on the particle, and which of them does work?

STAGE 2 Simple harmonic motion in s

Ask yourself Use $s = 2\sqrt{ky}$ to replace y by a function of s. What kind of motion does the resulting equation describe?

STAGE 3 The time to the bottom

Pause and try Why does this work? The "restoring force" along the curve is g times the slope, $g\frac{\mathrm{d}y}{\mathrm{d}s} = \frac{g}{2k}s$: exactly proportional to the distance from the bottom, measured along the curve. That is the defining property of simple harmonic motion. A particle starting higher has further to go, but is pushed proportionally harder.

STAGE 4 Two particles

Ask yourself When do the particles reach V, and how fast are they moving? Who has more momentum?

Ask yourself With total momentum zero and coefficient of restitution α, what are the velocities just after the collision?

Try this next

1 **Different heights.** A particle is released from height h in the tautochrone bowl. Show that its speed at V is $\sqrt{2gh}$, as it must be by energy conservation, and check that this agrees with ωs_0.

2 **An ordinary bowl.** In a hemispherical bowl of radius R, compare the time taken to reach the bottom from a small height with the time from the rim. Which is longer? (Use the result of Section 11.3.)

3 **Equal masses.** Repeat the second part of the question with two particles of equal mass m, released from heights h_1 and h_2, and coefficient of restitution e. Do they still always collide at V?

4 Cycloid arc length. Show that one arch of a cycloid has length $8R$, and that the area under it is $3\pi R^2$, three times the area of the rolling circle.

Further reading

- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press). Includes the tautochrone and the brachistochrone, with their history.
- **E. A. Whitman, "Some historical notes on the cycloid"**, *American Mathematical Monthly* 50 (1943). Why the cycloid was called "the Helen of geometers".
- **Paul Nahin, *When Least Is Best*** (Princeton University Press). The brachistochrone and other optimisation problems, told with history.

The full guided solution, and what the question leads to at university, is in Section 11.4:
https://beyondstep.keithmcnulty.org/ch11/tautochrone.html